D.Y. PATIL INTERNATIONAL UNIVERSITY

PYTHON MINOR PROJECT

PASSWORD GENERATOR

A Dark-Themed GUI Application Using Python Tkinter

Submitted by	Ganesh Ishwar Bhoge
PRN Number	26013020231
Branch	AIML
Batch	Batch-3
Class / Division	IBM-2
College	D.Y. Patil International University
Subject	Python Programming

Academic Year: 2026–2027

1. Introduction

The Password Generator is a Python desktop application that creates random passwords based on the user's preferences. Users can choose password length and include uppercase letters, lowercase letters, digits and special symbols. The application has a dark-themed graphical user interface (GUI).

2. Objectives

• Understand Python GUI programming using Tkinter.
• Generate random passwords using Python libraries.
• Customize password length and character types.
• Use secure random selection through Python's secrets module.
• Provide a copy-to-clipboard feature.

3. Software and Hardware Requirements

Software: Python 3.x, Tkinter, and a code editor such as IDLE or VS Code.
Hardware: Computer/laptop with a supported operating system and Python installed.

4. Modules Used

tkinter: Creates the GUI controls.
messagebox: Displays warning and information pop-ups.
secrets: Selects characters using a cryptographically strong random source.
string: Provides predefined character groups.

5. Working of the Project

1. The application opens and displays password settings.
2. The user chooses a length from 4 to 64 characters.
3. The user selects character categories.
4. On clicking Generate Password, at least one character from each selected category is chosen.
5. Remaining characters are selected from the combined pool and shuffled.
6. The password is displayed; the Copy button copies it to the clipboard.

6. Algorithm

Start → Initialize GUI → Read length and selected groups → Validate selection → Pick one character from each selected group → Fill remaining positions → Securely shuffle → Display password → Wait for next action → Exit.

7. Security Features and Limitations

The project uses Python's secrets module rather than the ordinary random module for password character selection. It ensures every selected category appears at least once. The application does not save passwords to a file or send them over the internet. Users should use unique passwords for accounts and store them in a trusted password manager.

8. Testing

Test case	Input / Action	Expected result
1	Generate with all options selected	All selected categories appear
2	Uncheck every category	Warning is displayed
3	Length too short for selected groups	Warning is displayed
4	Copy before generating	Information message appears
5	Generate and click Copy	Password copied to clipboard

9. Conclusion

The Password Generator demonstrates Python programming, GUI design, input validation and secure random character selection. It is a useful beginner-level project with scope for further improvements.

10. Future Scope

Future improvements may include a password strength meter, custom symbol sets, generating multiple passwords and password-manager integration.

11. References

• https://docs.python.org/3/
• https://docs.python.org/3/library/tkinter.html
• https://docs.python.org/3/library/secrets.html
• https://docs.python.org/3/library/string.html